Physics 468/668: Statistical Physics **Spring 2019**

Lecture 19: Ising model

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

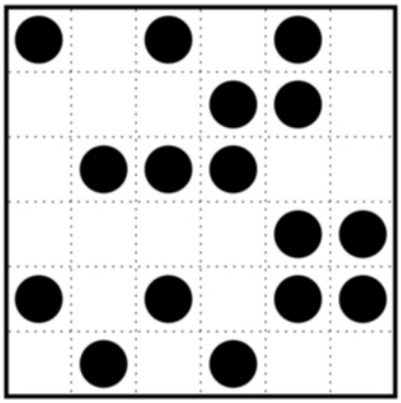


Figure 19.1: Lattice model.

19.1 A lattice Model

A number of physico-chemical systems that undergo phase transitions can be represented, to varying degrees of accuracy, by an *array of lattice sites, with only nearest-neighbor interaction that depends on the manner of occupation of the neighboring sites*. This can explain a variety of phenomena such as ferromagnetism and antiferromagnetism, gas-liquid and liquid-solid transitions, order-disorder transitions in alloys, phase separation in binary solutions, and so on.

The Hamiltonian of the system in configuration $\{\sigma_1, \sigma_2, \cdots \sigma_N\}$ is then given by

$$H\{\sigma_i\} = -J \sum_{N.N.} \sigma_i \sigma_j - \mu B \sum_i \sigma_i \tag{19.1}$$

where J is the coefficient related to two spin configurations, and each σ_i takes -1 or +1 configuration.

The partition function is

$$\begin{aligned} Q_N(B,T) &= \sum_{\sigma_1}\sum_{\sigma_2}\cdots\sum_{\sigma_N} \exp[-\beta H\{\sigma_i\}] \\ &= \sum_{\sigma_1}\sum_{\sigma_2}\cdots\sum_{\sigma_N} \exp\left[\beta J \sum_{N,N} \sigma_i\sigma_j + \beta\mu B \sum_i \sigma_i\right] \end{aligned} \tag{19.2}$$

The Helmholtz free energy, the internal energy, the specific heat, and the net magnetization of the system then follow from the formulae

$$\begin{aligned} A(B,T) &= -kT \ln Q_N(B,T) \\ U(B,T) &= -T^2 \frac{\partial}{\partial T}\left(\frac{A}{T}\right) = kT^2 \frac{\partial}{\partial T} \ln Q_N \\ C(B,T) &= \frac{\partial U}{\partial T} = -T\frac{\partial^2 A}{\partial T^2} \end{aligned} \tag{19.3}$$

and

$$\overline{M}(B,T) = \overline{\mu\left(\sum_i \sigma_i\right)} = \overline{\left(-\frac{\partial H}{\partial B}\right)} = \frac{1}{\beta}\left(\frac{\partial \ln Q_N}{\partial B}\right)_T = -\left(\frac{\partial A}{\partial B}\right)_T \tag{19.4}$$

The quantity $\bar{M}(0,T)$ gives the spontaneous magnetization of the system. If it is nonzero at temperatures below a certain critical temperature T_c, the system would be ferromagnetic for $T < T_c$ and paramagnetic for $T > T_c$. At the transition temperature itself, the system is expected to show some sort of a singular behavior.

The energy depends only on a few numbers such as the total number N_+ of up spins, the total number N_{++} of upup nearest-neighbor pairs, and so on. To see this, we define certain other numbers as well: N_- as the total number of down spins, N_{--} as the total number of downdown nearest-neighbor pairs, and N_+ as the total number of nearest-neighbor pairs with opposite spins. The numbers N_+ and N_- must satisfy the relation

$$N_+ + N_- = N \tag{19.5}$$

If q denotes the coordination number of the lattice, we have

$$\begin{aligned} qN_+ &= 2N_{++} + N_{+-} \\ qN_- &= 2N_{--} + N_{+-} \end{aligned} \tag{19.6}$$

With the help of these relations, we can express all numbers in terms of any two of them.

$$N_- = N - N_+, \quad N_{+-} = qN_+ - 2N_{++}, \quad N_{--} = \frac{1}{2}qN - qN_+ + N_{++} \tag{19.7}$$

It will be noted that the total number of pairs of all types is given,

$$N_{++} + N_{--} + N_{+-} = \frac{1}{2}qN \tag{19.8}$$

Naturally the Hamiltonian of the system can also be expressed in terms of N_+ and N_{++}; we have

$$\begin{aligned} H_N(N_+, N_{++}) &= -J(N_{++} + N_{--} - N_{+-}) - \mu B(N_+ - N_-) \\ &= -J(\frac{1}{2}qN - 2qN_+ - 4N_{++}) - \mu B(2N_+ - N) \end{aligned} \tag{19.9}$$

Now, let $g_N(N_+, N_{++})$ be "the number of distinct ways in which the N spins of the lattice can be so arranged as to yield certain preassigned values of the numbers N_+ and N_{++}. The partition function of the system

can then be written as

$$Q_N(B,T) = \sum_{N_+,N_{++}}{}' g_N(N_+,N_{++}) \exp[-\beta H_N(N_+,N_{++})] \tag{19.10}$$

So,

$$Q_N(B,T) = e^{-\beta A} = e^{-\beta N(\frac{1}{2}qJ-\mu B)} \sum_{N_+=0}^{N} e^{-2\beta(qJ-\mu B)N_+} \sum_{N_{++}}{}' g_N(N_+,N_{++}) e^{4\beta J N_{++}} \tag{19.11}$$

where the primed summation goes over all values of N_{++} that are consistent with a fixed value of N_+ and is followed by a summation over all possible values of N_+, that is from $N_+ = 0$ to $N_+ = N$. The central problem thus consists in determining $g_N(N_+, N_{++})$ for the various lattices of interest.

19.2 Lattice gas

A lattice gas is a collection of atoms, N_a in number that can occupy only discrete positions in space - positions that constitute a lattice structure with coordination number q. Each lattice site can be occupied by at most one atoom, and the interaction energy between two occupied sites is nonzero, say $-\epsilon_0$, only if the sites involved constitute a nearest neighbor pair. The configurational energy of the gas is

$$E = -\epsilon_0 N_{aa} \tag{19.12}$$

where N_{aa} is the total number of nearest-neighbor pairs in a given system. Let $g_N(N_a, N_{aa})$ denote "the number of distinct ways in which the N_a atoms of the gas, assumed indistinguishable, can be distributed among the N sites of the lattice so as to yield a certain preassigned value of N_{aa}." THe partition function of the system, neglecting the kinetic energy of the atoms, is given by

$$Q_{N_a}(N,T) = \sum_{N_{aa}}{}' g_N(N_a,N_{aa}) e^{\beta\epsilon_0 N_{aa}} \tag{19.13}$$

where the primedd summation goes over all values of N_{aa} that are consistent with the given values of N_a and N. Clearly, N here is analogical to volume available to the lattice gas.

Going over to the grand canonical ensemble, we write the paritition function

$$\mathcal{Q}(z,N,T) = \sum_{N_a=0}^{N} z^{N_a} Q N_a(N,T) \tag{19.14}$$

The pressure P and the mean number $\bar{N_a}$ of the atoms in the gas is

$$e^{\beta PN} = \sum_{N_a=0}^{N} z^{N_a} \sum_{N_a a}{}' g_N(N_a,N_{aa}) e^{\beta\epsilon_0 N_{aa}} \tag{19.15}$$

and

$$\frac{\bar{N_a}}{N} = \frac{1}{v} = \frac{z}{kT}\left(\frac{\partial P}{\partial z}\right)_T \tag{19.16}$$

The lattice gas	The ferromagnet
$N_a, N - N_a$	$N_+, N_- N_+ (= N_-)$
$-\epsilon_0$	$4J$
z	$\exp[-2\beta(qJ - \mu B)]$
P	$-\left(\frac{A}{N} + \frac{1}{2}qJ - \mu B\right)$
$\frac{\bar{N}_a}{N}$	$\frac{\bar{N}_+}{N}\left(= \frac{1}{2}\left[\frac{\bar{M}}{N\mu} + 1\right]\right)$

Table 19.1: The relation between lattice gas and binary alloy.

where

$$\bar{M} = \mu(\bar{N}_+ - \bar{N}_-) = \mu(2\bar{N}_+ - N) \tag{19.17}$$

$$\frac{\bar{N}_+}{N} = \frac{1}{kT}\left(\frac{\partial(A/N + 1/2qJ - \mu B)}{2\beta\partial(qJ - \mu B)}\right) = \frac{1}{2}\left[- \frac{1}{N\mu}\left(\frac{\partial A}{\partial B}\right)_T + 1\right] \tag{19.18}$$

which leads to

$$\frac{\bar{N}_+}{N} = \frac{1}{2}\left(\frac{\bar{M}}{N\mu} + 1\right) \tag{19.19}$$

Is this system related to any real physical system in nature? Note that this model automatically includes a short-range repulsion between the particles, since no two particles are allowed to share a same cell. The simplest instance of the lattice gas is obtained when $J = 0$, in which case the Hamiltonian is identically 0. Since its particles only interact through short-range repulsion, this model is called the hard-core lattice gas.

19.3 The Binary alloy

Much of the early activity in the theoretical analysis of the Ising model was related to the study of order-disorder transitions in alloys. In an alloy to be specific, a binary alloy we have a lattice structure consisting of two types of atoms, say 1 and 2, numbering N_1 and N_2, respectively. In a configuration characterized by the numbers N_{11}, N_{22}, and N_{12} of the three types of nearest-neighbor pairs, the configurational energy of the alloy may be written as

$$E = \epsilon_{11}N_{11} + \epsilon_{22}N_{22} + \epsilon_{12}N_{12} \tag{19.20}$$

As in the case of ferromagnet, the various numbers appearing in the expression for E may be expressed in terms of the numbers N, N_1, N_{11}, the equation can be expressed as:

$$\begin{aligned} E &= \epsilon_{11}N_{11} + \epsilon_{22}\left(\frac{1}{2}qN - qN_1 + N_{11}\right) + \epsilon_{12}(qN_1 - 2N_{11}) \\ &= \frac{1}{2}q\epsilon_{22}N + q(\epsilon_{12} - \epsilon_{22})N_1 + (\epsilon_{11} + \epsilon_{22} - 2\epsilon_{12})N_{11} \end{aligned} \tag{19.21}$$

The relation between this system and the lattice gas is now straightforward

The lattice gas	The binary alloy
$N_a, N - N_a$	$N_1, N - N_1 (= N_2)$
$-\epsilon_0$	$(\epsilon_{11} + \epsilon_{22} - 2\epsilon_{12})$
A	$A - \frac{1}{2} q \epsilon_{22} N - q(\epsilon_{12} - \epsilon_{22}) N_1$

Table 19.2: The relation between lattice gas and binary alloy.

At absolute zero, our alloy will be in the state of minimum configurational energy, which would also be the state of maximum configurational order. We expect that the two types of atoms would then occupy mutually exclusive sites, so that one might speak of atoms 1 being only at sites a and atoms 2 being only at sites b. As temperature rises, an exchange of sites results and, in the face of thermal agitation, the order in the system starts giving way. Ultimately, the two types of atoms get so mixed up that the very notion of the sites a being the right ones for atoms 1 and the sites b being the right ones for atoms 2 break down; the system then behaves, from the configurational point of view, as an assembly of $N_1 + N_2$ atoms of essentially the same species.